

Inputs

Image / Visual evidence

documents, charts, science diagrams, natural images

Question prompt

question text + optional answer choices

Target format

Answer only:

”Answer: <answer>”

Rationale target:

”Reasoning: <rationale> Answer: <answer>”

Qwen2-VL Branch

Qwen2-VL 2B / 7B

Visual encoder

frozen

Language decoder

4-bit NF4 frozen base

QLoRA adapters

trainable

attention + MLP layers

BLIP-2 Branch

BLIP-2 OPT-2.7B

ViT vision encoder

frozen

Q-Former

trainable in BLIP runs

32 learned query tokens

OPT language model

frozen

Qwen2-VL Supervision

Answer-only CE

train only answer tokens

Rationale-generative CE

train full rationale+answer

Explanation-aware CE

separate rationale span
and answer span

Loss

$L = \alpha * CE_{answer} + (1-\alpha) * CE_{rationale}$

BLIP-2 Supervision

Generative CE

rationale + answer targets

Contrastive-enhanced

generative CE + InfoNCE image-answer loss

Image embedding

pooled Q-Former queries

Text embedding

pooled answer tokens projected to Q-Former dim

Model Outputs

Predicted answer

Generated rationale when required

Image-answer embeddings for BLIP-2 retrieval check

Evaluations

Correctness:

MC accuracy, ANLS, relaxed accuracy, VQA accuracy

Rationale quality:

ROUGE-L, BLEU on rationale datasets

Grounding:

image masking drift / evidence sensitivity

Transfer:

source adapter evaluated on other domains